MILITARY SPECIFICATION

CARTRIDGE, 20MM, ARMOR PIERCING, INCENDIARY - TRACER, MARK 161 MOD 0

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of percussion primed cartridge for use in Guns, 20mm, Automatic. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition
MIL-STD-651	- Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY ARMAMENT COMMAND

F7258942	- Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258969	- Method of Packing 100 Cartridges, 20mm, in M14 Type Links, in M548 Container

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2.1 (Cont)

D11075818	- Marking on M548 Shipping and Storage Container for 100 20mm Cartridges in M14 Type Links
D11075819	- Marking on M548 Shipping and Storage Container for 200 20mm Cartridges in Bulk Pack
D11075822	- Cartridge, 20mm, Target Practice-Tracer, MK 161 Mod 0
IEL 11075822	- Index of Inspection Equipment Lists for: Cartridge, 20mm, Target Practice-Tracer, MK 161, Mod 0

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall be in accordance with Drawing (Dwg.) D11075822, referenced specifications, and other requirements specified herein.

3.2 First Article Sample.-Unless otherwise directed by the contracting officer, a First Article sample shall be required.

3.3 Projectile Extraction.-The force necessary to extract the projectile from the cartridge case shall be as specified on Dwg. D11075822.

3.4 Velocity.-The average velocity of the sample cartridges conditioned at 68 degrees (°) Fahrenheit (F) to 72°F shall be 3380 ± 50 feet per second (fps) at 78 feet from the test weapon muzzle and the sample standard deviation shall not exceed 40 fps.

3.5 Pressure.-The average chamber pressure ($\overline{X}$) of the sample cartridges, conditioned at 68°F to 72°F, plus 0.84 times the average range ($\overline{R}$) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

3.6 Waterproofness.-The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F to 72°F shall not vary from the average velocity (dry) by more than 100 fps.

3.7 Accuracy.-The average mean radius of all targets of the sample cartridge projectiles at ambient temperature shall not exceed 15 inches at a range of 600 yards.

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3.8 Action Time.-The action time of the cartridge at ambient temperature shall not exceed 4.0 milliseconds. The action time of the sample cartridges conditioned at -65°F shall not exceed 4.5 milliseconds.

3.9 Projectile Trace.-The projectile shall exhibit a visible trace after a minimum of 0.8 seconds and continue to have a visible trace for 3.0 seconds minimum after cartridge ignition when viewed against a dark background.

3.10 Function and Casualty.-The cartridge, at ambient temperature, shall function without casualty.

3.11 Workmanship.-The requirements for workmanship are as specified in the applicable drawing, referenced specifications, and the following:

3.11.1 Process Defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, cracks, splits, severe scratches, cuts, dents, perforations, and other forms of substandard workmanship.

3.11.2 Cleanliness.-The cartridge and its components shall be free of all foreign matter. (see 6.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3).
2. Quality Conformance Inspection (see 4.4).

4.3 First Article Inspection.

4.3.1 First Article Sample.-At the beginning of production, a sample representative of the production material, equipment, processes and procedures shall be submitted in accordance with contract requirements and shall consist of 440 cartridges. Prior to approval of the first article sample, the acquisition of material or components for, or the commencement of production of the balance of the contract quantity shall be at the sole risk of the contractor.

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4.3.1.1 First Article Inspection.-Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings and specification requirements. The contractor's inspection shall be subjected to verification by the Government representative.

4.3.1.2 Examination and Tests.-The first article sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such facility specified in the contract. Determination as to approval of any first article sample shall be based only upon results of initial examinations and tests.

4.4 Quality Conformance Inspection.

4.4.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the contract, the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.4.1.2 Lot Identification.-Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a National stock number assigned by the procuring activity.

4.4.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

4.4.2.1 Sampling and Acceptance Criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.1.1.c) In addition, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.3)

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4.4.2.1.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

4.4.3 Classification of Defects 1/.-The classification of defects shall be as follows:

4.4.3.1 Projectile (see Dwg. D-LC-SK-948, a detail of Dwg. D11075822)

Category and Defects	Method of Inspection	
Critical: None defined.		
Major:		
101. Missing or improper metering disc	Visual	
102. Metering disc crimp defective	Visual/manual	2/
103. Seal missing or perforated	Visual	
104. Foreign matter	Visual	
Minor: None defined.		

4.4.3.2 Cartridge Assembly (see Dwg. D11075822)

Category and Defect	Method of Inspection	
Critical:		
1. Split or perforated case	Visual	
2. Crack or split in projectile	Visual	
Major:		
101. Loose, cocked, crushed, inverted or missing primer	Manual/visual	
102. Depth of primer seating	Gage	
103. Over-all length, max.	Gage	
104. Improper, incomplete, or missing crimp (case-projectile)	Visual	
105. Profile and alignment, max.	Gage	3/
106. Corrosion	Visual	
107. Head configuration	Visual	

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4.4.3.2 (Cont)

Minor:

201. Marking incorrect, incomplete, illegible or missing --- Visual
202. Projectile protective coating damaged or incomplete --- Visual
203. Waterproofing missing around primer ---------- Visual
204. Scratch, dent, buckle, bulge, wrinkle or fold in case ---------- Visual
205. Rotating band damaged ---------- Visual
206. Primer crimp missing ---------- Visual
207. Foreign matter, except corrosion ---------- Visual
208. Incorrect type cartridge or component ---------- Visual
209. Dented or deformed nose ---------- Visual

1/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

2/ If the crimp closure is not pressed tightly against the metering disc for 360° or the metering disc can be moved manually, the charged body subassembly shall be classed defective.

3/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

4.4.4 Tests.-The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. The sample cartridges shall be selected in such a manner that the sample is representative of the entire cartridge lot. The cartridge lot sample shall be examined in accordance with 4.4.2 and 4.4.3 and from this sample the cartridges for the various test quantities shall be selected at random. (see 6.4)

Table I

Test 1/		Sample Size Production Lot	Rqmt Para
Projectile extraction	2/	20	3.3
Propellant Contamination	3/	20	3.11.2
Velocity	4/	20	3.4
Pressure	5/	20	3.5
Waterproofness	6/	20	3.6
Accuracy	7/	40	3.7
Action time	8/		3.8
Ambient temperature		25	
-65°F temperature		25	
Projectile trace	9/	50	3.9
Function and casualty	10/	200	3.10

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4.4.4 (Cont)

1/ The lot shall be rejected if in any firing test one or more of the following malfunctions or casualties of Table II occur:

a. Projectile remaining in bore.

b. Metal parts separation, except rotating band separation (see Table II, Note 4/).

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

2/ Projectile Extraction.

a. Failure of two or more sample cartridges to conform with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to conform with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to conform with the minimum requirement.

b. Failure of four or more cartridges to conform with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to conform with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to conform with the maximum requirement.

3/ Propellant Contamination.-The presence of foreign matter in the propellant of one or more of the test cartridges will cause rejection of the lot. (see 3.11.2 and 6.5)

4/ Velocity.-If the sample fails to conform with either or both requirements, a retest sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to conform with the applicable requirements. (see 6.6.1)

5/ Pressure.-If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 60,500 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to conform with the applicable requirement. (see 6.6.2)

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4.4.4 (Cont)

6/ Waterproofness.-If the average velocity (wet) of the sample fails to conform with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to conform with the requirement of 3.4.

7/ Accuracy.-If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 80 cartridges shall be tested. The lot shall be rejected if the average mean radius of eight accuracy targets exceeds 15 inches at the same range.

8/ Action Time.-If any sample cartridge fails to meet the requirements for action time, the lot shall be rejected.

9/ Projectile Trace.-Failure of 9 or more cartridges of the sample to comply with the requirements shall cause rejection of the lot. If more than 5 and less than 9 sample cartridges fail to comply with the requirements, a second sample of 50 cartridges shall be tested. If, in the combined first and second sample, 13 or more sample cartridges fail to comply with the requirements, the lot shall be rejected.

10/ Function and Casualty.-The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

4.4.4.1 Firing Defects.-Firing defects and associated criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

Table II

Firing Defects	Acc	Rej
Cartridge:		
Misfire	0	2
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1

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4.4.4.1 (Cont)

Firing Defects	Acc	Rej
Primer:		
Blown primer	0	1
Loose primer	0	2
Primer perforation	1	6
Primer leak	9	25
Case:		
Longitudinal split 1/:		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture (partial) 1/:		
S, J, or K	4	11
L	0	1
Circumferential rupture (complete) 1/	0	1
Detached metal 2/	0	1
Projectile:		
Rotating band separation:		
Complete or partial separation having band seat serration imprint 3/	0	3
Separation having length greater than 0.5 inch without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 4/	0	1

1/ Dwg. C7258801 (a detail of IEL 11075822) shall apply for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

2/ Metal sheared or missing from the fired cartridge case exterior, such as a rim or neck shear, shall be classed as a defect. The lot shall not be penalized for metal shaved from the interior surface of the case neck in the crimped area.

3/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of a band fragment having no evidence of a band seat knurling imprint, and having length of 0.5 inch or less.

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4.4.4.1 (Cont)

4/ Separation or breakup of projectile part(s), except rotating band separation, as witnessed by recovery of the part(s), of fragment(s), or hole(s) in the fragmentation screen(s), shall be classed as a defect.

4.4.4.2 Unlisted Firing Defects.-The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test indicates that the product is unsuited for the purpose intended.

4.4.5 Packaging, Packing, and Marking Inspection.-Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644, as applicable.

4.4.6 Inspection Equipment.-Index of Inspection Equipment List No. IEL 11075822 identifies the applicable Inspection Equipment List (IEL) required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. B11075228, a detail of IEL 11075822).

4.5 Test Methods and Procedures.

4.5.1 Projectile Extraction.-The method of test shall be as specified in AMCR 715-505, Volume 8.

4.5.2 Propellant Contamination.-The propellant from each of the cartridges tested for projectile extraction shall be examined visually for contamination, as specified in AMCR 715-505, Volume 8.

4.5.3 Velocity, Pressure and Waterproofness.-The methods of test shall be as specified in AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed. (see 6.6)

4.5.4 Accuracy.-The method of test and measurement of targets shall be as specified in AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.5.5 Action Time.-The method of test shall be as specified in AMCR 715-505, Volume 8. The sample cartridges shall be fired single shot.

4.5.6 Projectile Trace.-The method of test shall be as specified in AMCR 715-505, Volume 8, except that observation shall be from near the gun position and timing shall be from cartridge ignition to the start of visible trace and to the end of visible trace.

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4.5.7 Function and Casualty.-The method of test shall be as specified in AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges. The weapon barrel shall be at ambient temperature at beginning of test and cooled to ambient after each 100 rounds. The cyclic rate of each burst shall not be less than 475 shots per minute (SPM) in the MARK 29 gun. The cyclic rate for each burst shall be recorded.

4.5.8 Defective Weapon.-If a test weapon fails to conform with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst, the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. Any firing defect that occurred shall be valid for lot penalty in accordance with Table II unless the defect is found to have been caused by the defective weapon.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with instructions from the procuring activity. Linked ammunition shall be packed in accordance with Dwg. F7258969. Bulk ammunition shall be packed in accordance with Dwg. F7258942.

5.2 Marking.-Marking on the shipping and storage container shall be in accordance with Dwg. D11075818 or D11075819 as applicable.

6. NOTES

6.1 Intended Use.-These cartridges are intended for use in various gas operated automatic gun systems that have been designed for firing percussion primed cartridges with the M103 type case configuration.

6.2 Ordering Data.-Invitation for bids and contracts or orders will specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 First article sample requirements. (see 3.2 and 4.3)

6.2.4 Facility for the inspection of the first article sample. (see 4.3.1.2)

6.2.5 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.2.6 Provision for the supply, maintenance and disposition of Government Furnished test equipment for acceptance inspection purposes.

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6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 Combining Tests.-With approval of the procuring activity, tests may be performed concurrently on the same sample cartridges provided that the test results are not affected by this procedure. The test facility should make maximum use of this procedure in order to minimize testing costs.

6.5 Contamination of Explosive Components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfires, increase action time and lot rejection.

6.6 Computations.

6.6.1 Standard Deviation.-Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$S = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n}}$$ or equivalent.

6.6.2 Range Plan.-Where a range plan is specified, the individual values will be recorded in subgroups of five in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range. $(\bar{R})$.

Custodians:
Army - MU
Navy - OS

Preparing Activity:
Army - MU

Review Activities:
Army - MU
Navy - OS

Project No. 1305-0779

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